

PROFESSIONAL SUMMARY:

- Data & AI/ML Engineer with 3+ years of experience architecting end-to-end data ecosystems across Data Engineering, Data Science, AI/ML Engineering, and Analytics, a true full-stack data professional comfortable owning a problem from raw ingestion to executive dashboard.
- Architected and delivered enterprise-grade Azure data platforms (ADF, Synapse Analytics, Databricks, ADLS Gen2), engineering high-throughput ETL/ELT pipelines in Python, SQL, and PySpark that transformed raw, unstructured data into analytics-ready assets at scale.
- Built and deployed production machine learning systems, from phishing detection to KYC document intelligence, using Scikit-learn, XGBoost, and LLM-based techniques, with rigorous evaluation (Accuracy, Precision, Recall, F1) and experiment tracking via MLflow and MLOps.
- Designed scalable data models (fact/dimension schemas) and optimized storage architecture (Parquet on ADLS Gen2), enabling serverless querying at scale through Synapse SQL Pools.
- Translated complex technical findings into business impact through Power BI and Tableau dashboards, partnering directly with cross-functional stakeholders to drive data-informed decision-making.
- Combined technical depth with Agile leadership as a Certified Scrum Master, facilitating sprint ceremonies, removing blockers, and mentoring teammates to improve delivery velocity, bridging the gap between engineering execution and business strategy.

TECHNICAL SKILLS:

Category	Key Tools & Concepts
Programming Languages	Python, SQL, PySpark
Azure Data Engineering	Azure Databricks, Azure Data Factory (ADF), Azure Synapse Analytics, ADLS Gen2, Azure SQL Database, Synapse Serverless SQL Pools, Azure DevOps, CI/CD pipelines, Azure Key Vault, Azure Monitor, Log Analytics, Docker, Delta Lake
Data Engineering & Processing	ETL/ELT pipelines, data ingestion, cleaning & transformation, data modeling, Parquet, Pandas, NumPy, Apache Spark, feature engineering, Data governance, data quality frameworks, Unity Catalog (Databricks), Star schema / Snowflake, schema design, Batch & streaming pipelines
Databases	Azure SQL Database, PostgreSQL, MySQL, SQL Server, MongoDB
Generative AI	Prompt Engineering, RAG (Retrieval-Augmented Generation), LangChain, LangGraph, LLM, Embeddings, Vector Database, Deep Learning, NLP, CNN, RNN, FastAPI, Azure OpenAI Service
Machine Learning	Supervised & Unsupervised Learning, XGBoost, LightGBM, Random Forest, SVM, K-Means, PCA, Ensemble Methods, Anomaly Detection, LLM techniques, Scikit-learn, MLflow, MLOps, Model deployment, model monitoring, drift detection
Visualization	Matplotlib, Seaborn, Plotly, Power BI, Tableau
Version Control	GitHub

EDUCATION:

- Master of Science in Computing and Data Science
- Master of Business Administration

PROFESSIONAL EXPERIENCE:

Freelancing
Machine Learning Engineer

Apr 2025 - PRESENT

Project: AI-Powered KYC Document Intelligence & Verification System

Tech Stack: Azure Databricks, PySpark, Python, Azure, LLM, Agile/Scrum, MLFlow, MLOps

Project Overview: Developed an end-to-end production-grade system for automated KYC document processing to classify, validate, and extract insights from identity documents such as passports, driver's licenses, and financial records.

Roles and Responsibilities:

- Built a data pipeline to ingest and store unstructured KYC documents (PDFs/images) in Azure Data Lake, enabling scalable and secure storage for raw document data.
- Engineered a production-grade KYC document intelligence pipeline, combining computer vision-adjacent document classification with LLM-driven extraction, reducing manual document review effort and accelerating verification turnaround.
- Processed and cleaned document data using Python, extracting relevant fields such as name, ID number, and document type for further analysis.
- Designed and developed an end-to-end AI application from initial architecture through cloud deployment using Azure, Python, and LLM technologies.
- Built scalable AI pipelines supporting data ingestion, document preprocessing, model inference, and automated information extraction.
- Developed RESTful inference services using FastAPI to expose AI capabilities for enterprise document processing.
- Integrated embedding-based semantic retrieval techniques into LLM workflows to improve contextual understanding and response quality.
- Deployed production-ready AI services with MLflow-based model tracking, version management, and performance monitoring.
- Containerized AI components using Docker to support consistent deployment across development and production environments.
- Collaborated with cross-functional stakeholders to deliver scalable AI solutions aligned with business requirements.
- Developed basic classification models in Python to categorize documents (e.g., passport, invoice, ID), improving document organization and processing efficiency.
- Integrated LLM-based processing to generate summaries and extract key insights from documents, supporting faster verification and decision-making.
- Deployed the solution on Azure cloud services, ensuring scalability and smooth integration between storage, processing, and model components.
- Implemented CI/CD pipelines using Azure DevOps to automate AI application deployment, testing, and release management.
- Supported production AI applications by monitoring system health, resolving deployment issues, and implementing performance improvements.
- Optimized machine learning inference pipelines and data processing workflows to improve scalability, response time, and resource utilization.
- Designed modular cloud-native AI applications using Azure services to support scalable, secure, and highly available enterprise solutions.
- Worked in an Agile/Scrum environment, contributing to sprint planning and task tracking while collaborating with team members to deliver features on time.

Ananvay Consulting LLC, Las Vegas, NV

Aug 2024 – Mar 2025

Data Engineer and Scrum Master

Project: Vehicle Complaints Analytics

Tech Stack: Azure Databricks, PySpark, Azure Data Factory (ADF), Azure Synapse Analytics, Azure Data Lake Storage Gen2, SQL, Python, Parquet, Serverless SQL Pools, Power BI, Data Modeling, Agile/Scrum

Project Overview: Built a cloud-based data pipeline on Azure to process and analyze vehicle complaints data. Focused on creating a simple, scalable workflow from data ingestion to reporting dashboards.

Key Responsibilities

- Data Ingestion: Used Azure Data Factory (ADF) to ingest raw vehicle complaints data from source systems into Azure Data Lake Storage Gen2, handling basic scheduling and pipeline automation.
- Data Storage & Format Optimization: Stored data in Parquet format in ADLS Gen2 to improve storage efficiency and enable faster querying for large datasets.
- Owned the full data lifecycle for a vehicle complaints analytics platform, from ADF-orchestrated ingestion through Synapse-powered serverless querying to Power BI storytelling, while simultaneously driving Agile delivery as Scrum Master, balancing technical execution with team leadership.
- Data Cleaning & Transformation: Used SQL and Python to clean data (remove nulls, duplicates) and transform it into structured formats suitable for analysis.

- **Data Processing & Querying:** Created external tables using Azure Synapse Serverless SQL Pools to query data directly from the data lake without moving it.
- **Data Modeling:** Designed simple fact and dimension tables to organize data for reporting and analytics.
- **Visualization & Reporting:** Connected data to Power BI and built dashboards to analyze complaint trends, categories, and locations.
- **Agile/Scrum Leadership:** Facilitated daily stand-ups and sprint planning, ensuring tasks were clearly defined and aligned with project goals.
- Collaborated with team members to track progress, remove basic blockers, and improve sprint delivery efficiency.

Evolent, Arlington, VA
Associate Data Scientist

Nov 2022 – Jun 2024

Project: Enterprise Phishing Detection System

Tech Stack: Python (Scikit-learn, XGBoost), SQL Server, MLflow, Tableau

Project Overview: Built a machine learning pipeline to detect phishing emails and malicious URLs using structured features and text data. Focused on building a simple, scalable workflow from data ingestion to model monitoring and visualization.

Key Responsibilities

- **Data Collection & Storage:** Collected phishing and safe email data, stored and managed datasets in SQL Server, and used SQL queries to extract, clean, and prepare data for analysis.
- Built and operationalized a phishing detection ML pipeline achieving reliable classification performance, using MLflow for full experiment lifecycle management and Tableau to translate model behavior into stakeholder-ready insights.
- **Data Preprocessing & Feature Engineering:** Performed data cleaning (handling nulls, duplicates) and created features such as URL length, presence of special characters, and TF-IDF vectors from email text using Python (Scikit-learn).
- Managed the complete machine learning lifecycle, including data preparation, feature engineering, model training, evaluation, deployment, and monitoring.
- Built production-grade ML pipelines capable of handling large-scale structured and text datasets using Python, SQL Server, and MLflow.
- Optimized feature engineering workflows and model performance through iterative experimentation and version-controlled model management.
- **Model Building & Evaluation:** Trained classification models including Logistic Regression and XGBoost, and evaluated performance using metrics like Accuracy, Precision, Recall, and F1-score to ensure reliable phishing detection.
- **Experiment Tracking & Model Management:** Used MLflow to track experiments, compare different models, and store model versions, helping in selecting the best-performing model.
- **Visualization & Insights:** Created interactive dashboards in Tableau to visualize model performance, feature importance, and phishing trends, enabling easy understanding for stakeholders.

Authorized to work in the USA (EAD holder). No visa sponsorship is required. Open to relocation.

CERTIFICATIONS:

- CompTIA IT Fundamentals
- Scrum Master Certified